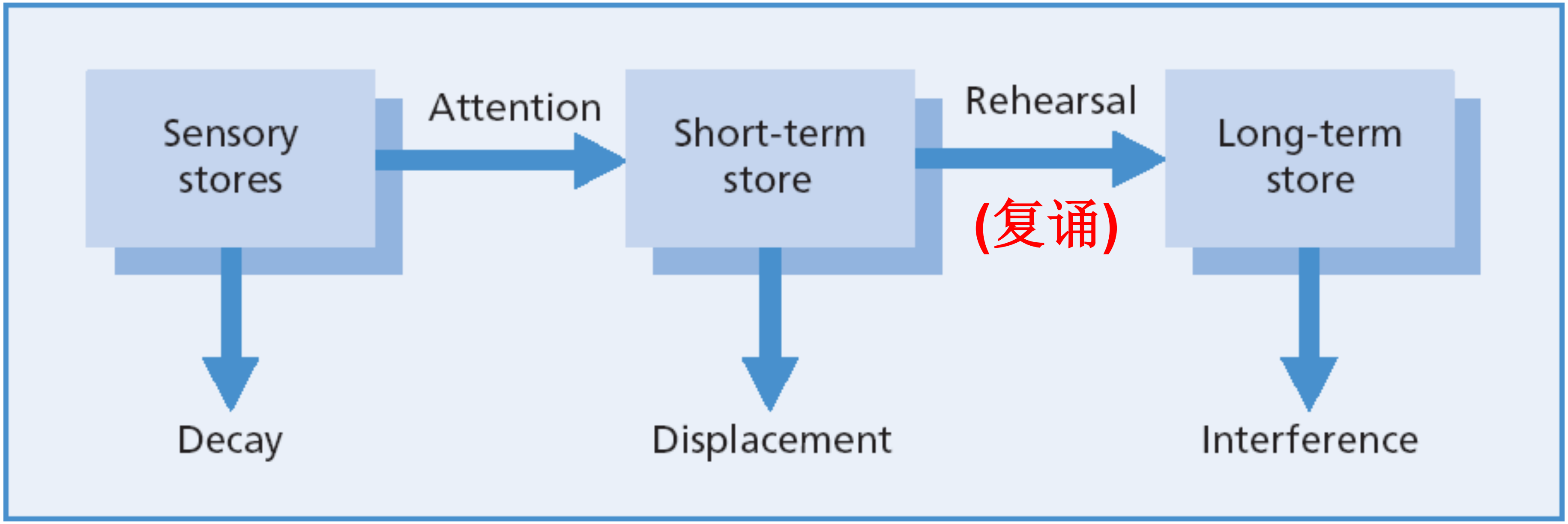
Memory III

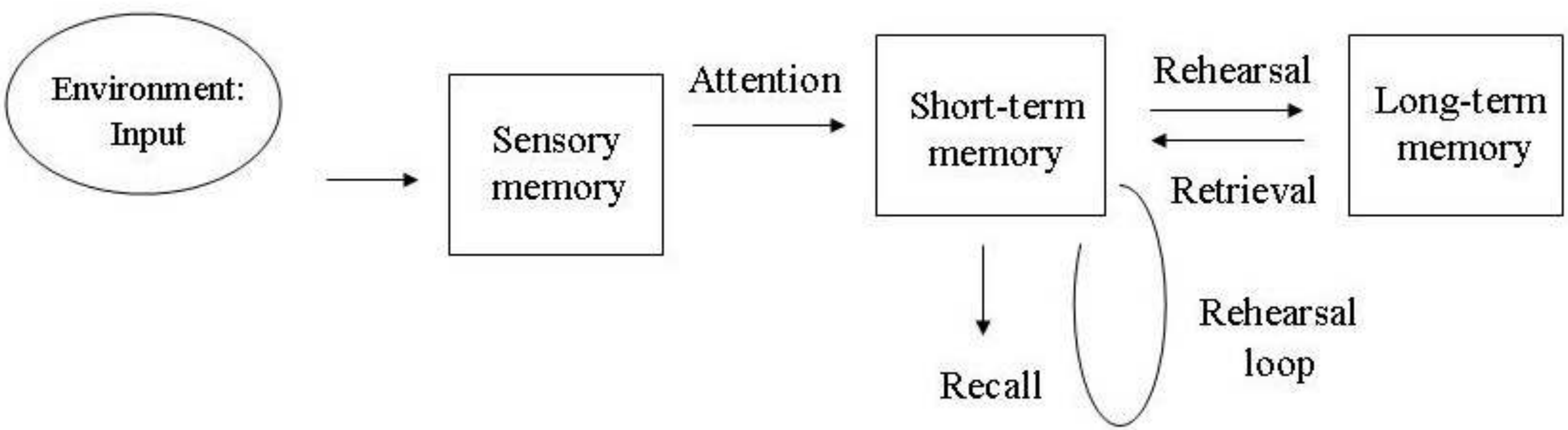
Working Memory & Brain

Atkinson & Shiffrin (1968) Model of Memory



Atkinson & Shiffrin (1968) Model of Memory

- **Multi Store Model (Atkinson and Shiffrin, 1968)**
- **The multi store model** is a classic model of memory. It is sometimes called the **modal model** or the **dual process model**
- **Atkinson and Shiffrin (1968)** suggest that memory is made up of a series of stores (see below)

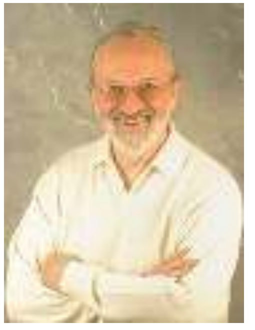


Visual Sensory Store

- It appears that our visual system is able to hold a great deal of information but that if we do not attend to this information it will be rapidly lost.

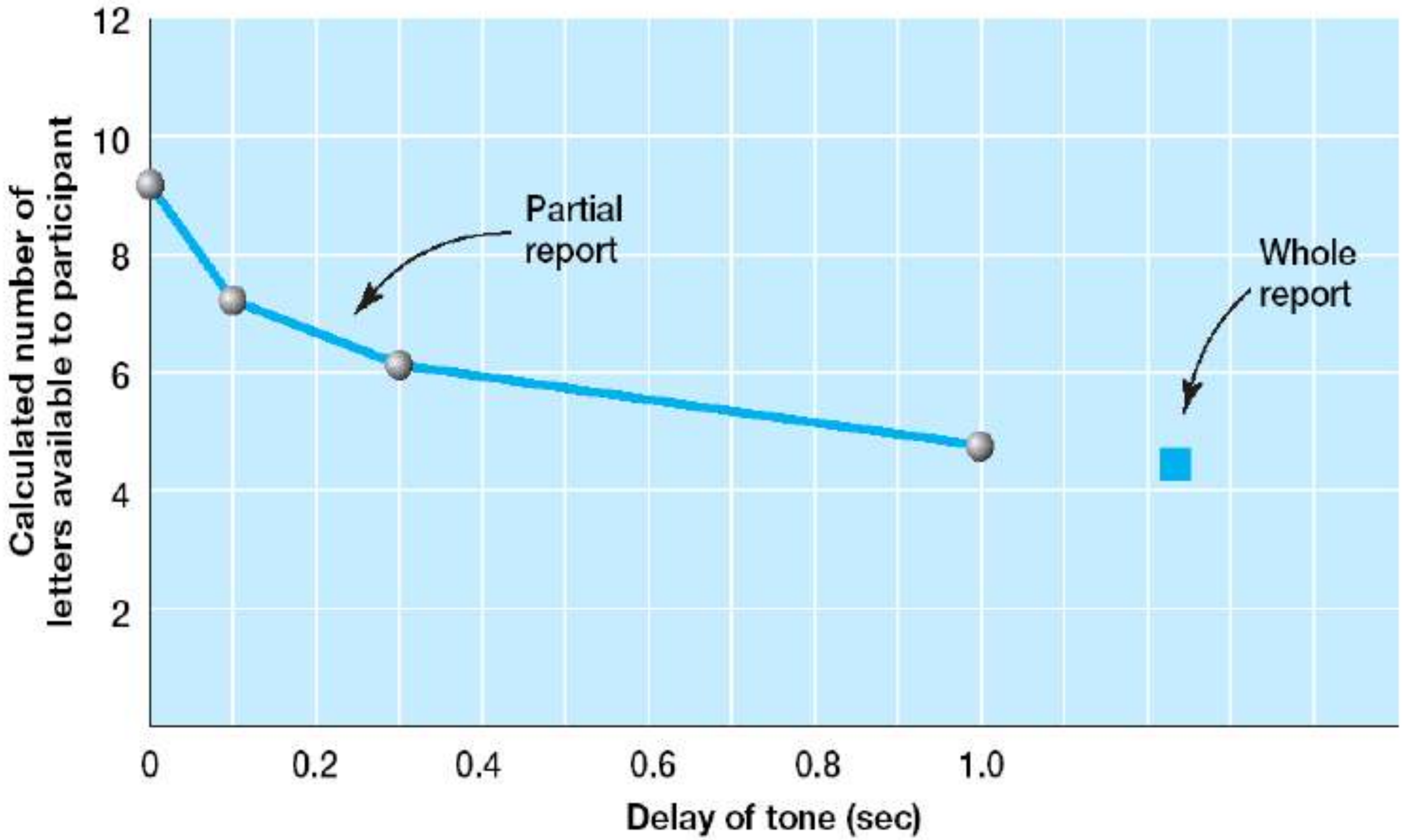
X	**M**	**R**	**J**
C	**N**	**K**	**P**
V	**F**	**L**	**B**

- Sperling (1960)
  - Presented array consisting of three rows of four letters
  - Subjects were cued to report part of or whole display



Demo at: http://www.dualtask.org/

Visual Sensory Memory



Iconic memory ➔ high capacity, rapid decay

Iconic Memory

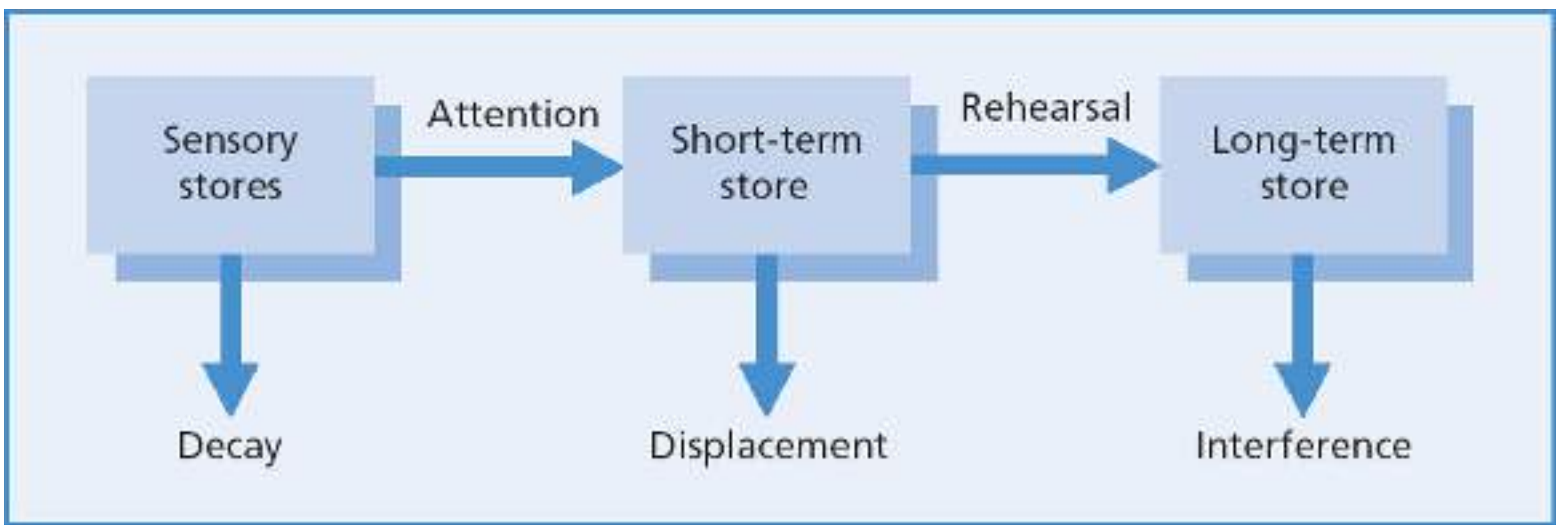
- Sperling's experiments indicate the existence of a brief visual sensory memory -- known as **iconic memory** or **iconic store**

- Information decays rapidly (after a few hundred milliseconds) unless attention transfers items to short-term memory

- Analogous auditory store: **echoic store**

Atkinson & Shiffrin (1968) Model of Memory

- **Short-term memory** (STM) is a **limited capacity** store for information -- place to rehearse new information from sensory buffers
- Items need to be **rehearsed** in short-term memory before entering long-term memory (LTM)
- Probability of encoding in LTM directly related to **time** in STM



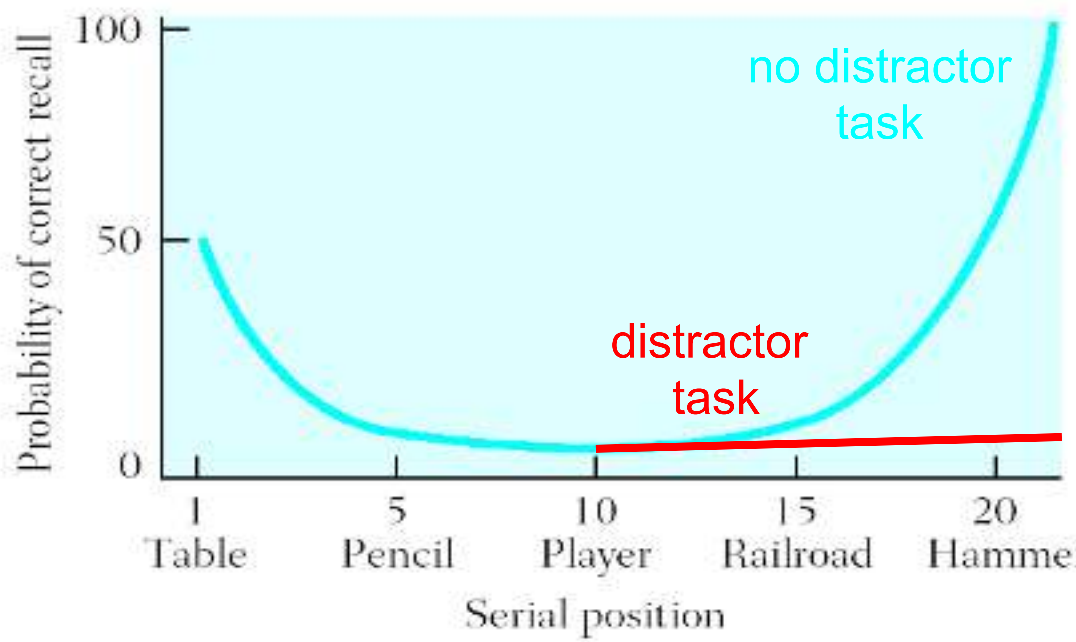
a memory test...

Serial Position Effects (序列位置效应)

- In free recall, more items are recalled from start of list (primacy effect) (首因效应) and end of the list (recency effect) (近因效应)
- Distractor task (e.g. counting) after last item removes recency effect

1 Table	11 Kitten
2 Candle	12 Doorknob
3 Maple	13 Folder
4 Subway	14 Concrete
5 Pencil	15 Railroad
6 Coffee	16 Doctor
7 Towel	17 Sunshine
8 Softball	18 Letter
9 Curtain	19 Turkey
10 Player	20 Hammer

(A)



(B)

Serial Position Effects

- Explanation from Atkinson and Shiffrin (1968) model:

  - Early items can be rehearsed more often

  ➔ more likely to be transferred to long-term memory

  - Last items of list are still in short-term memory (with no distractor task)

  ➔ they can be read out easily from short-term memory

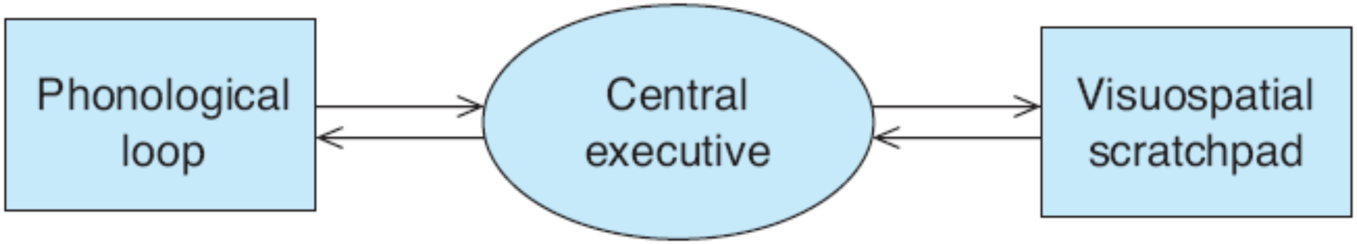
Atkinson & Shiffrin (1968) Model of Memory

Evaluating Modal Memory Model

- Pro
  - provides good quantitative accounts of many findings
- Contra
  - assumption that all information must go through STM is probably wrong
  - Model proposes one kind of STM but evidence suggests we have multiple kinds of STM stores

Baddeley's working memory model

- Baddeley proposed replacing unitary short-term store with working memory model with multiple components:



(语音回路)　(中枢执行系统)　(视觉空间模板)

- ➢ Key components:
- ➢ Central Executive; other two systems are slave systems.
- ➢ Phonological Loop holds information in phonological or speech-based form.
- ➢ Visuospatial sketchpad specialized for spatial and/or visual coding.

Allen Baddeley

Baddeley and Hitch (1974)
Baddeley (1986)

Phonological Loop

(a.k.a. articulatory loop)

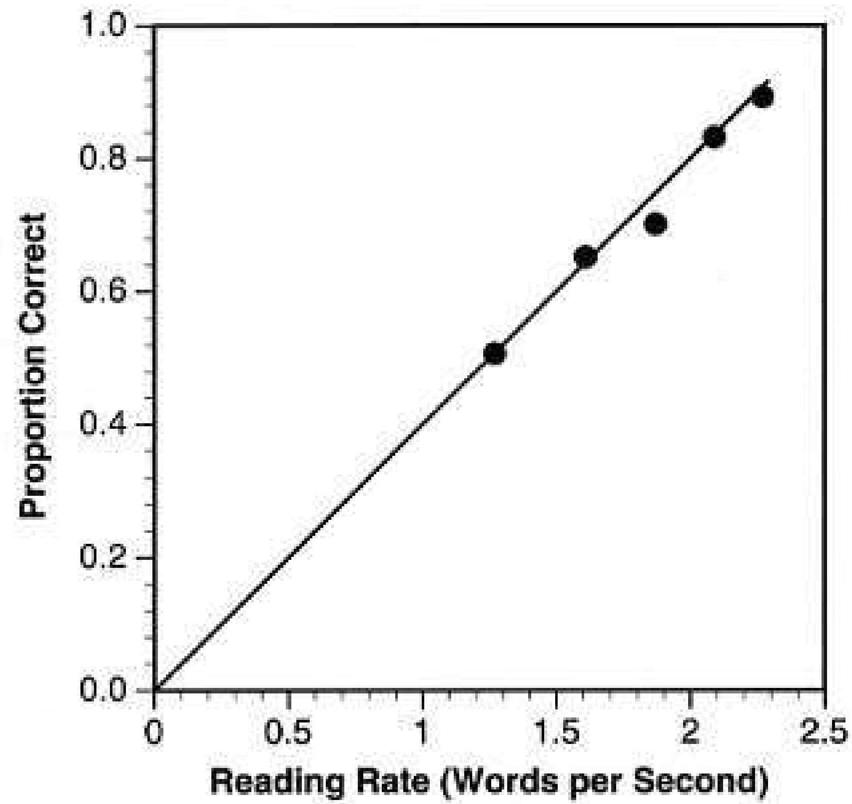
- Stores a limited number of sounds – number of words is limited by pronunciation time, not number of items
- Experiment:

LIST 1:	LIST 2:
Burma	Switzerland
Greece	Nicaragua
Tibet	Afghanistan
Iceland	Venezuela
Malta	Philippines
Laos	Madagascar

- Word length effect -- mean number of words recalled in order (list 1 ➔ 4.2 words; list 2 ➔ 2.8 words)

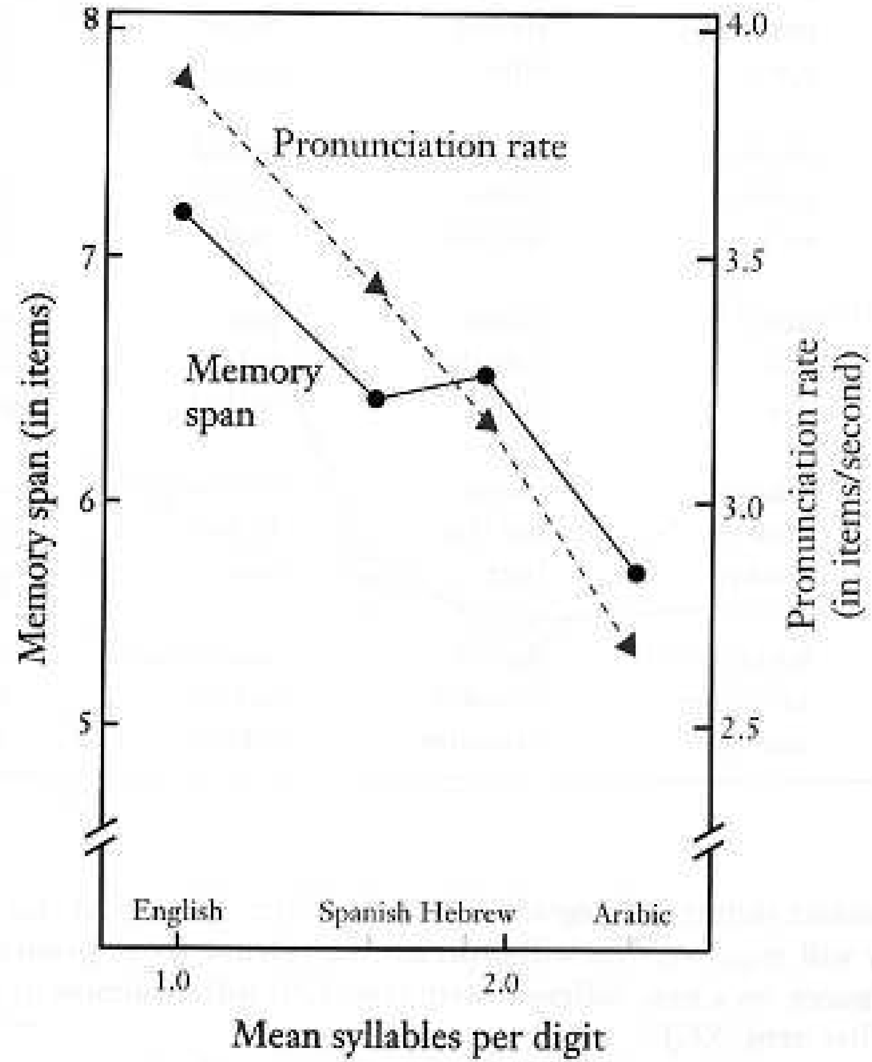
Reading rate determines serial recall

- Reading rate seems to determine recall performance
- Phonological loop stores 1.5 - 2 seconds worth of words



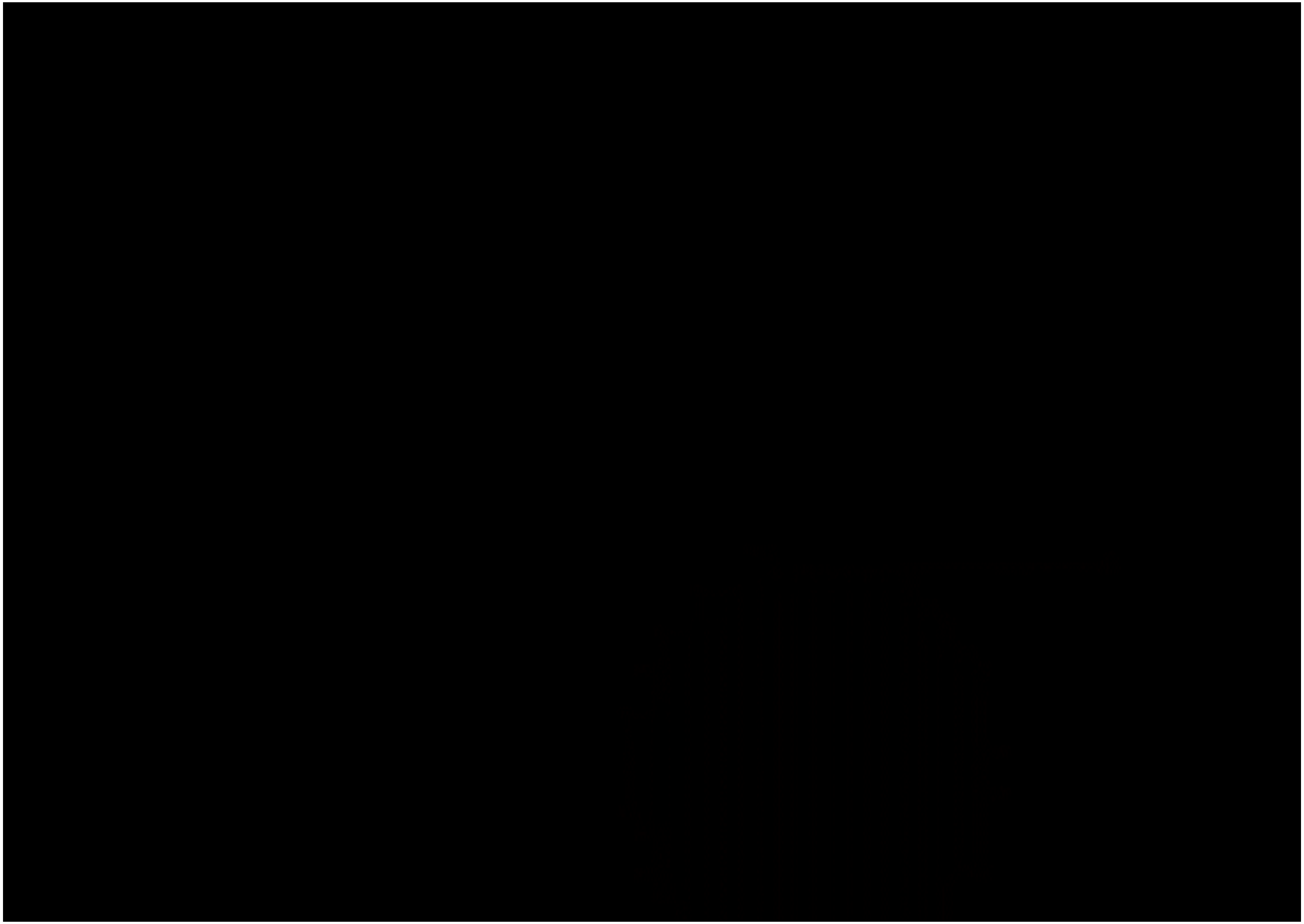
Working memory and Language Differences

- Different languages have different #syllables **(音节)** per digit
- Therefore, recall for numbers should be different across languages
- E.g. memory for English number sequences is better than Spanish or Arabic sequences



(Naveh-Benjamin & Ayres, 1986)

The working memory model

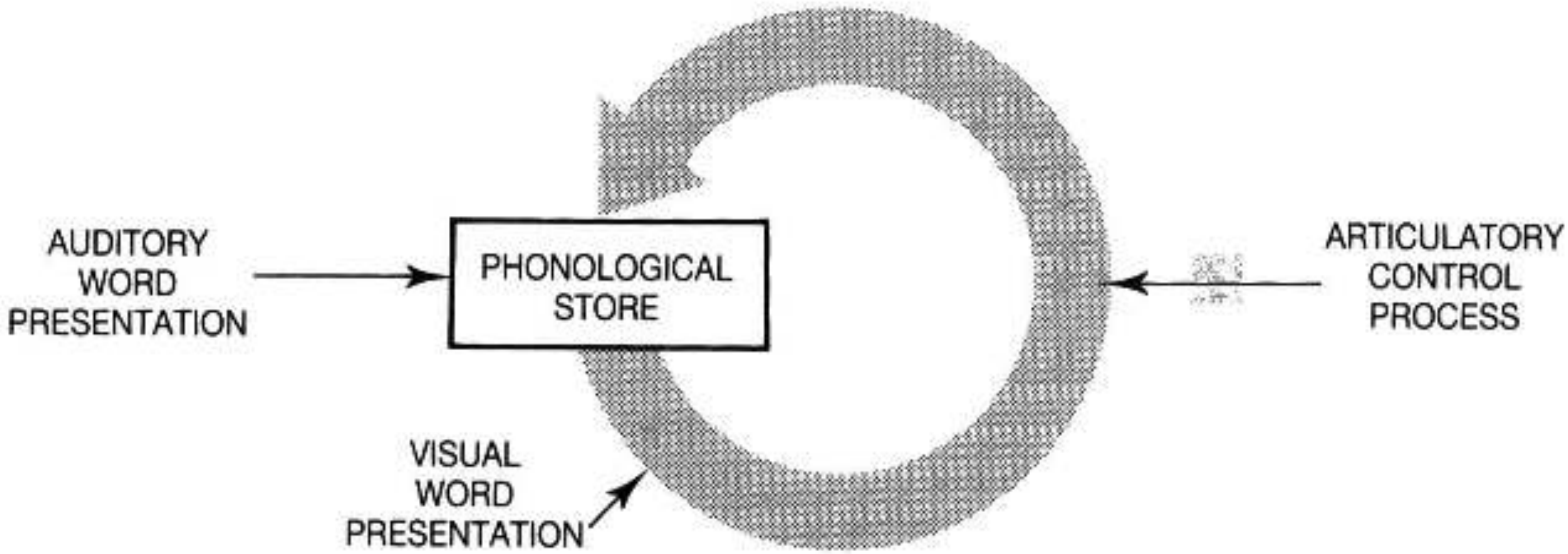


Features of the Phonological Loop

- Phonological store
  - Auditory presentation of words has direct access
  - Visual presentation only has indirect access
  - Affected by phonological similarity

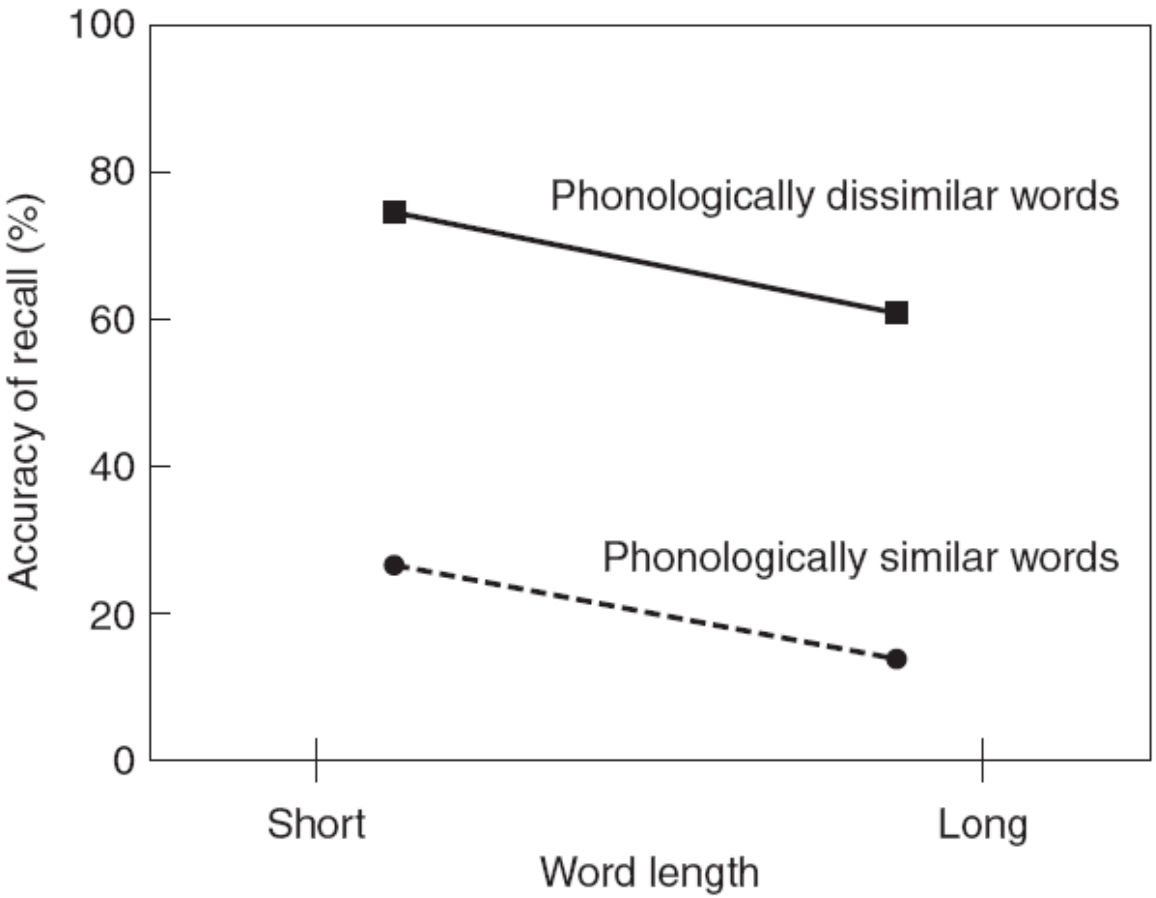
- Articulatory process
  - Converts visually presented words into inner speech that can be stored in phonological store
  - Affected by word length

Articulatory Suppression



By auditory rehearsal, a representation in the phonological store can be maintained

Articulatory Suppression



Articulatory Suppression

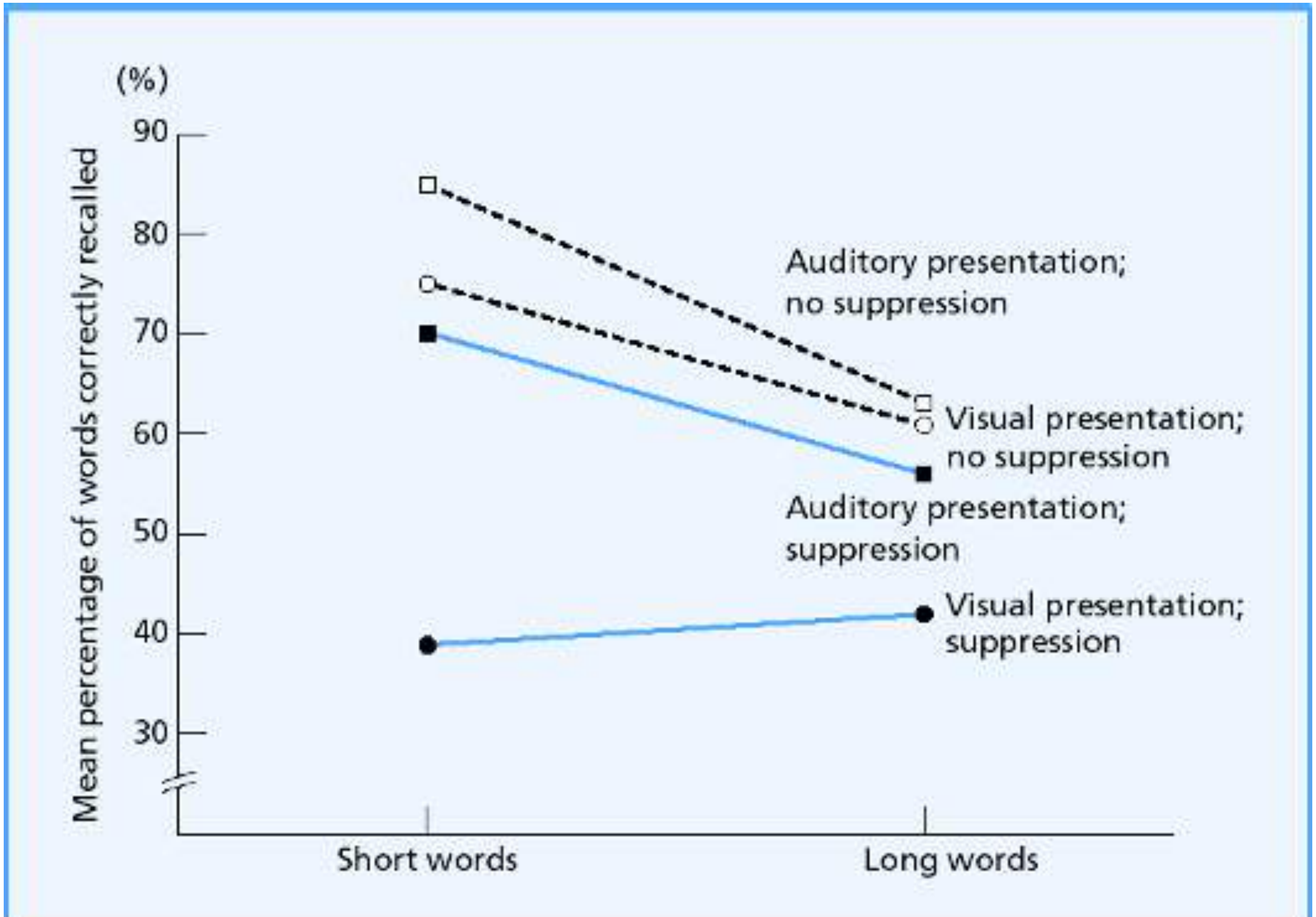
- Saying “the” all the time leads to **articulatory suppression**

- Disrupts phonological loop ➔ worse performance

- With visual presentation, articulatory suppression leads to bad performance but there is <u>no</u> word length effect

  ➔ visuospatial sketchpad takes over

Articulatory Suppression



Immediate word recall as a function of modality of presentation (visual vs. auditory), presence vs. absence of articulatory suppression, and word length.

Baddeley et al. (1975)

Neural Network Models of Memory

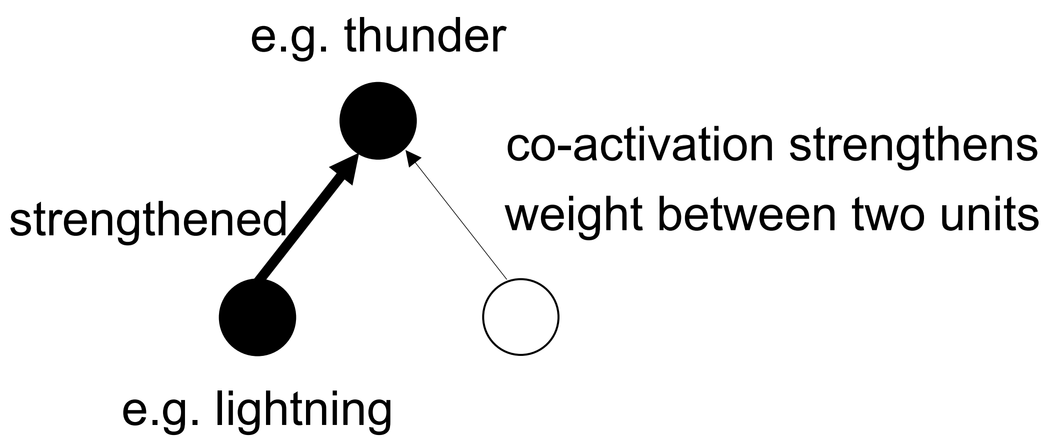
Neural Network Models of Memory

- Long-term memory:
  - weight-based memory
  - the memory representation takes its form in the strength or weight of neural connections

- Short-term memory:
  - activity-based memory
  - in which information is retained as a temporary (短暂的) pattern of activity in specific neural populations

Long-term memory

- Long-term associative memories can be formed by Hebbian learning: changes in synaptic (突觸) weights between neurons
  - structural change
  - relatively permanent
- Long Term Potentiation (LTP) is the biological basis of Hebb’s learning rule

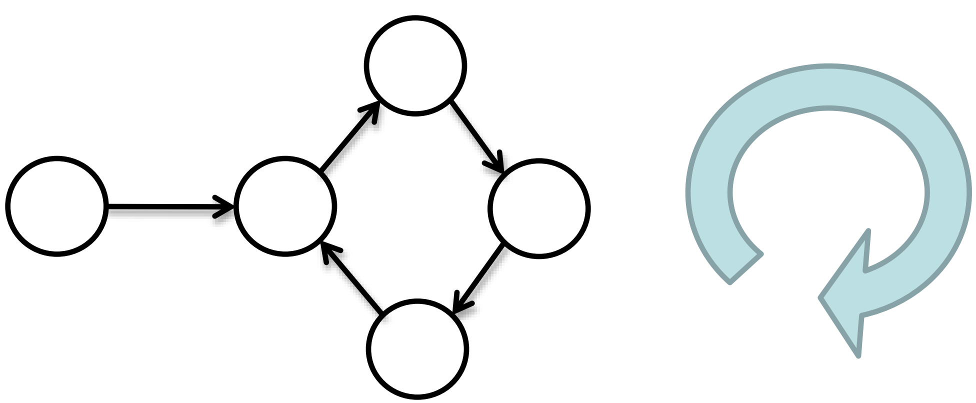


Donald O. Hebb

Short-term Memory

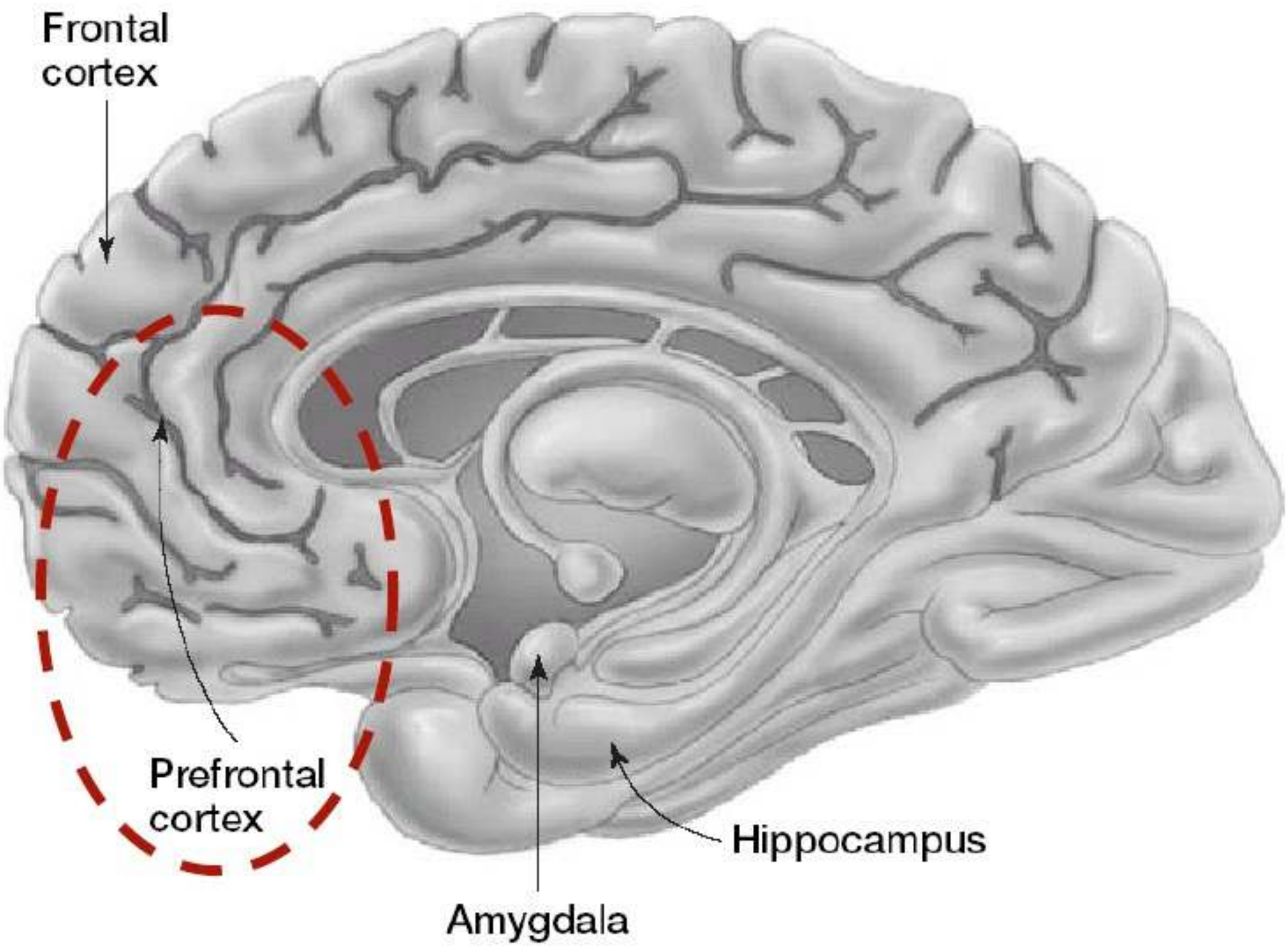
- Change in neural activity
  - ➢ not structural
  - ➢ temporary

- **Reverberatory loop (反射回路)** -- circuits that maintain activity for a short period



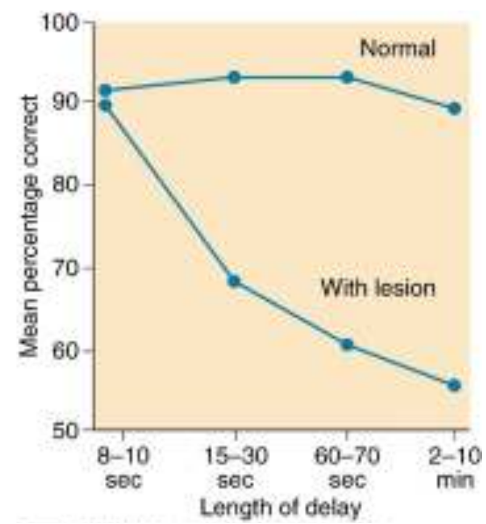
Working Memory and Prefrontal Cortex

(PFC: 前额皮质)



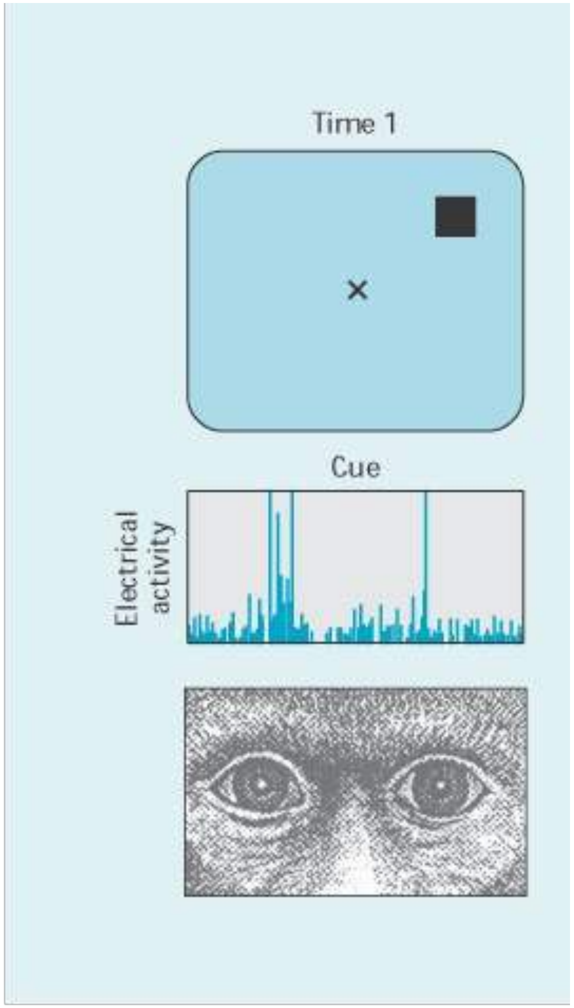
Delayed Match to Sample Tasks

- Correct response requires keeping location of food in mind
- Monkeys and humans with lesions of PFC fail these tasks



Delayed Saccade Task

(Goldman-Rakic)



Patricia Goldman-Rakic (1937-2003)

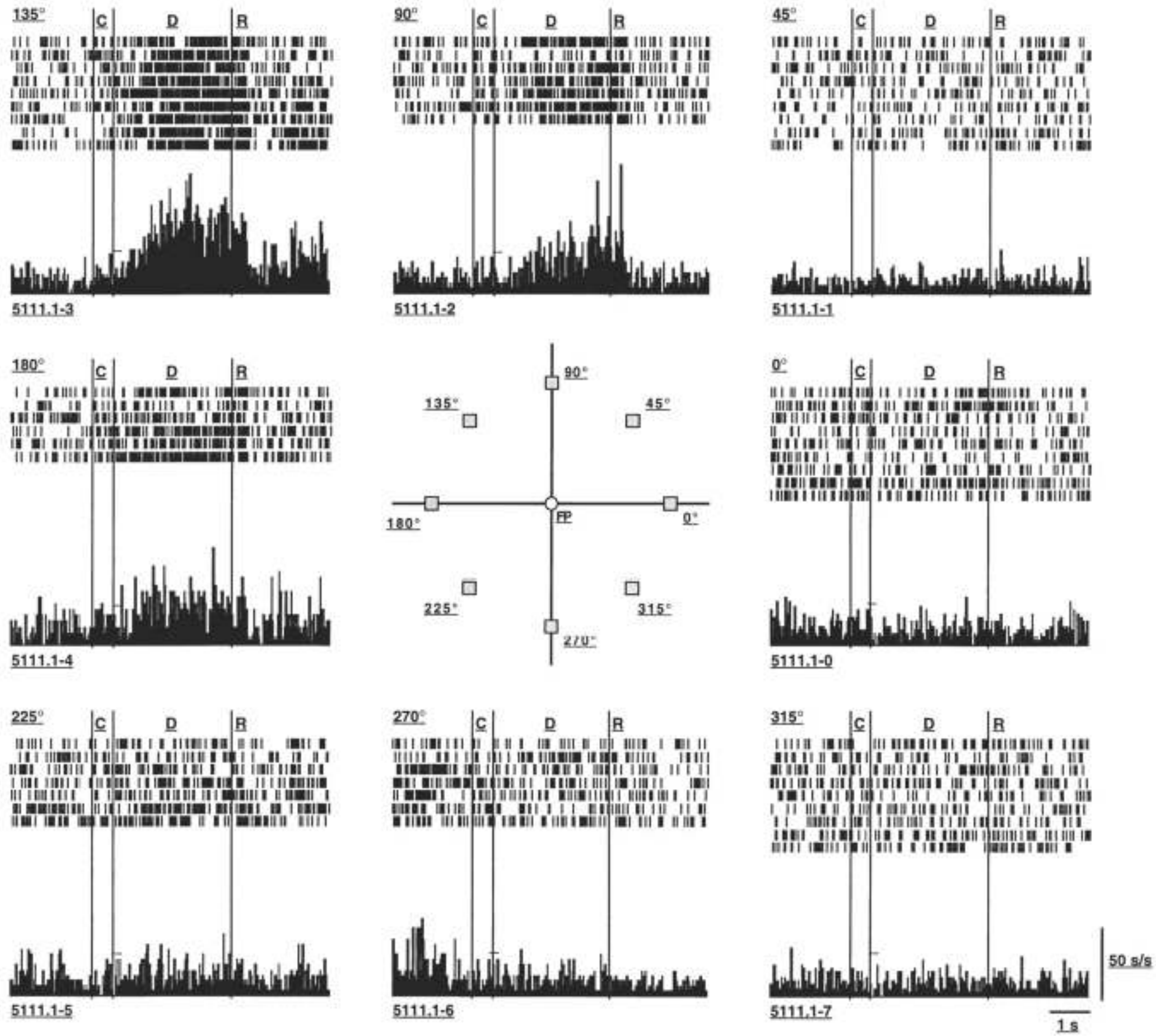


Fig. 2. Activity of a single prefrontal neuron, exemplifying persistent discharges during the execution of the oculomotor delayed-response task. Discharges are arranged as to indicate the location of the cue. The neuron is mostly active during the delay period following presentation of a stimulus in the upper left (135-degree) location. From Funahashi and others (1989) with permission.

Role of PFC in Memory Encoding

- If fMRI activity at encoding is back-sorted according to whether words are subsequently remembered or forgotten, then lower left VLPFC (and hippocampus) activation predicts later forgetting